

Randy Pahang II

(702) 609-6580 | rpahang2@gmail.com | linkedin.com/in/randypahangii | github.com/randyp2

EDUCATION

University of Nevada, Las Vegas

B.S. Computer Science — Honors College

Expected May 2027

GPA: 4.0/4.0

Relevant Coursework: Computer Networks, Cloud Computing, Data Structures and Algorithms, Linear Algebra, Machine Learning, Database Management

TECHNICAL SKILLS

Languages: C/C++/C#, Java, Python, TypeScript/JavaScript, SQL

Frameworks/Technologies: Spring Boot, FastAPI, Next.js, React, Vue, RabbitMQ, Redis, PostgreSQL, DynamoDB, Docker, AWS ECS, EC2, Lambda, S3, CloudFront, CloudWatch, CI/CD, YOLOv8, OpenAI API, Linux, Git, GCP

WORK EXPERIENCE

JT4, LLC

Software/Systems Engineering Intern — Telemetry Team

May 2026 – Present

Las Vegas, NV

- Built a real-time IRIG-106 Chapter 10 telemetry simulator in C++, eliminating an estimated **\$30K** in hardware costs by letting engineers validate data pipelines in-office instead of deploying to live test ranges
- Engineered packet generation streaming Chapter 10 data at live flight-test rates, enabling engineers to validate ingestion tools before consuming costly range time
- Built a PySide6/Qt dashboard surfacing real-time visibility into **100+** simulated telemetry channels for live stream-health monitoring
- Validated simulator output against recorded range data to drive adoption across telemetry teams' test pipelines

STARS Solution LLC

Software Engineering Intern

Nov 2025 – Present

Las Vegas, NV

- Built the Spring Boot backend for a veterans' career-guidance platform on **AWS ECS Fargate**, scaling stateless services for a **3.5K+ user** pilot
- Moved LLM plan generation off the request path with an async RabbitMQ pipeline, streaming partial results via Server-Sent Events instead of blocking waits
- Cut institution-search latency **3x (110ms to 37ms)** with a cache-aside Redis layer using versioned invalidation and graceful DB fallback
- Reduced irrelevant memory retrieval in offline evaluations by routing user information to structured or semantic storage and reranking context before LLM generation

PROJECTS

PortRN

Software Engineer (Founder)

Nov 2025 – Present

Las Vegas, NV

- Co-founded and launched an LLM-powered portfolio-generation platform with profile verification and social networking features for a **50+ user** beta cohort
- Cut generation time from **7–10 to 4–5 min** by parallelizing section generation across **10 RabbitMQ consumers**
- Reduced failed generations with a multi-stage JSX validator, including a sandboxed runtime render test, that routes errors to an LLM repair loop
- Built an idempotent Stripe webhook pipeline with signature verification and database-backed deduplication to prevent duplicate fulfillment across subscription and one-time payment flows

HotelGuard

Next.js, YOLOv8, Supabase

Jan 2026

1st Place, 35+ Teams — RebelHacks 2026

- Built a real-time human-trafficking detection system for hotels using YOLOv8 for human tracking, Gemini 2.5 for video analysis, and Next.js and Supabase for a live staff-monitoring dashboard
- Engineered a people-counting module using YOLOv8 object detection and 2D boundary-crossing logic to track room occupancy in real time and flag high-risk rooms
- Designed a weighted risk-scoring algorithm that aggregated occupancy patterns and behavioral signals into dynamic room-level threat scores visualized on a 3D React Three Fiber hotel floor plan